

PROTOCTISTS

From tiny amoebas to giant kelp, the protoctists defy simple description, yet this informal grouping of eukaryotes included the first life forms to evolve that were more complex than the archaea and bacteria. It still produces most of Earth's food and oxygen.

DOMAIN	EUKARYOTA
KINGDOM	PROTOCTISTA
CLADES AND DIVISIONS	More than 9
FAMILIES	About 841
SPECIES	About 73,500



In some slime moulds, which are a group of amoebas, thousands of individual nuclei coexist within one giant cell.



Many single-celled protoctists have extraordinary shapes, such as these grapple-like dinoflagellates.



Some protoctists cause devastating disease, such as this *Giardia*, which infects intestines of humans and other animals.

DEBATE

MULTIPLE KINGDOMS

The kingdom Protocista includes many organisms that are not classified as fungi, plants, or animals. And it includes many groups, ranging from single-celled amoebas to multicellular seaweeds, that are not closely related to one another. Many scientists think that Protocista should be split into more than one kingdom.

Protoctists are mainly single-celled creatures that, unlike archaea and bacteria, have cell nuclei. Their basic cell nature separates them from the higher eukaryotes – plants, fungi, and animals – that later emerged from them. The protoctists include an incredible range of organisms with diverse lifestyles and ecological niches. Most are microscopic, ranging in size from 10 to 100 μm , and some are tiny enough to infest red blood cells. Others, however, are multicellular organisms such as kelp, a seaweed that grows up to tens of metres in length, or the strange, fungus-like slime moulds, which form creeping puddles of slime that are essentially one giant cell. Typical protoctists include amoebas, which move about and capture food particles with their pseudopods (cell extensions), or plankton drifting in the sea, such as the beautiful diatoms, which have intricate silicon-based skeletons.

A HIDDEN REALM OF LIFE

Protoctists are among the most numerous creatures on Earth. Vast numbers of them live in the oceans and rivers, in sediments at the bottom of seas and lakes, and in the soil, while many spend all or part of their life cycles as parasites inside other organisms. They play a vital role in the planet's ecosphere, in particular as primary photosynthesizers, using the energy in light to convert carbon dioxide and water into food, while releasing oxygen into the atmosphere. They also act as predators and recyclers of matter. A few species are well known because they cause major diseases: the parasitic *Plasmodium* causes malaria, one of the biggest killers of humans; another parasite, *Trypanosoma brucei*, causes sleeping sickness. Also familiar are the dinoflagellates, a group of plankton organisms that can cause "red tides" – huge blooms of toxic creatures that kill fish and poison humans.

The taxonomy of the protoctists is complex as they don't form a natural kingdom. However, molecular and genetic analysis has allowed most protoctists to be placed in a few large clades, groups that each share a common ancestor, such as the Amoebozoa, Rhizaria, and Alveolata. The red algae, green algae, and the stoneworts each have their own division.

SMALL WONDERS >

This polarized light micrograph reveals an *Arcella* shelled amoeba and a *Micrasterias* desmid in all their beauty.

